Introduction

Uncontrolled hypertension represents the foremost risk factor for cardiovascular disease, making it the primary contributor to global all-cause mortality and morbidity in the United States.[1] Roughly half of US adults with uncontrolled high blood pressure have readings exceeding 140/90mmHg, amounting to approximately 37 million individuals with stage 2 hypertension.[2]

To address this pressing issue, it is suggested that improving self-management through education, dieting, and medication titration could be an effective way to reduce blood pressure in hypertensive patients, outside of in-person visits.

Notably, recent randomized controlled trials (RCTs) have demonstrated that telemonitoring for hypertension not only matches but often surpasses the blood pressure reductions achieved with usual care. It has shown superior efficacy in managing patients with uncontrolled hypertension compared to standard treatment methods.[3–6] Although RCTs suggest that home blood pressure monitoring, coupled with an intervention, can effectively lower blood pressure more than standard care, research beyond these trials remains limited.

As telehealth and virtual care become increasingly prevalent in healthcare practices and systems, addressing this knowledge gap is critical to developing an effective intervention to manage hypertension.

Telemonitoring demonstrates significant potential in managing hypertension by not only improving patient education and engagement with their health data but also enhancing the scale of monitoring. These tools enable clinicians to oversee a larger group of patients, providing timely alerts about patient status between visits. This approach also effectively channels limited staff resources towards the patients most in need of an intervention.

On the other hand, there are potential challenges associated with telemonitoring for blood pressure management. Patients may not be sufficiently engaged with the digital platform, which could affect their adherence to the prescribed management plan. Additionally, medications may require time to show their full effects, which could impact short-term outcomes measured by the platform.

Initial investigations into the feasibility of telemonitoring for hypertension management within primary care settings have yielded promising preliminary outcomes.[7]

Expanding upon this foundational research in practical environments, we implemented a hypertensive telemonitoring program and evaluated its impact on systolic blood pressure (SBP) and diastolic blood pressure (DBP) in stage 2 hypertensive patients.

Methods

Description of Intervention

Telemonitoring was conducted using an automated, text-message based chat program, termed a "virtual companion" in this study, designed to gather blood pressure readings and symptom reports from stage 2 hypertensive patients while also offering feedback and educational content. Patients in the program received weekly chat modules via text or email, and they also had the ability to opt out at any time during the pilot through these same channels.

The automated virtual companion prompted patients to input their blood pressure weekly, either manually or via a Bluetooth-enabled cuff. If the patient's last two blood pressure readings exceeded the threshold of 140/90 or were below 100/59 and the patient was asymptomatic, an alert was generated. Patients who received two alerts or reported symptoms, had their data sent via inbox message to the electronic medical record. These alerts were accessed by the central nursing pool who would triage the results and call patients to confirm their vitals and symptoms. The nursing team could then either direct patients to the appropriate level of care or forward a message to the patient's primary care physician, who would determine the need for interventions like medication titration or observation.

Study period and population

The study period spanned from April 2022 to February 2023. Patients eligible for inclusion were identified from an Adult hypertension Registry for Primary Care, based on having hypertension on their problem list and attending two or more outpatient visits for hypertension within the past two years. This criteria was refined further to include patients who met all of the following conditions: primary English speaking, ages 18 and above, having a primary care physician at our health system, and exhibiting uncontrolled stage 2 hypertension with a blood pressure reading of ≥140/90, as determined from their latest in-clinic ambulatory or self-reported blood pressure. Furthermore, patients recorded as pregnant were excluded from the pilot enrollment. Patients meeting the defined criteria were then extracted for enrollment, and a subset of patients activated their account and began to receive communications.

Estimating the impact of the virtual companion on blood pressure readings

The clinical goal of the pilot study was to evaluate the change in SBP and DBP within the population. For each patient, the time of their first collection was denoted as time 0, and readings thereafter were measured relative to the days elapsed since this initial reading. Change in blood pressure from baseline was then calculated at various times following this initial collection. For each of the following models discussed, separate models were created for SBP and DBP.

A linear mixed-effects model was fitted that regressed the days since a patient's first collection on change in blood pressure, with patient included as a random effect to account for multiple testing (referred to as Model 1 in this study).

In order to assess the intervention's effect on the target group of hypertensive individuals, patients were categorized into specific blood pressure classifications based on their initial reading:[8]

- Normal BP: Systolic less than 120 mmHg AND Diastolic less than 80 mmHg.
- Elevated BP: Systolic between 120-129 mmHg AND Diastolic less than 80 mmHg.
- Stage 1 hypertension: Systolic between 130-139 mmHg OR Diastolic between 80-89 mmHg.
- Stage 2 hypertension: Systolic 140 mmHg or higher OR Diastolic 90 mmHg or higher.

A second linear mixed-effects model added this classification to the model, along with an interaction between the classification and time (referred to Model 2 in this study). This allows for the assessment of blood pressure changes over time within classification. The ‘Normal BP’ classification was used as the reference category for this analysis.

Sensitivity analysis

In order to assess robustness of the model, a sensitivity analysis was performed that added other predictors (patient age, race, and the hour of blood pressure recording) to Model 2. A Likelihood Ratio Test comparing this model with Model 2 was performed to assess the combined significance of these additional predictors.

Results

Description of study population

A total of 665 patients were enrolled into the pilot and received an initial activation link. Among the 665 patients, 308 (46%) successfully activated their account with the chatbot and began to receive the weekly chat modules. Additionally, out of the activated patients, 161 (52%) patients recorded at least one blood pressure measurement during the course of the study. Of those who recorded a blood pressure reading, 81% were White and 15% were Black or African American, while 26% were under 65 years old, 31% between 65 and 75 years old, and 43% over 75 years old. This group had 12 patients (7%) with normal BP, 24 patients (15%) with elevated BP, 92 patients (57%) with stage 2 hypertension, and 33 patients (21%) with stage 2 hypertension.

The impact of the virtual companion on blood pressure readings

Across the entire patient population of those who recorded blood pressures, we observed a statistically significant reduction in both SBP and DBP (Model 1). Specifically, for each month (i.e. 30 days) following a patient's initial blood pressure recording, SBP decreased by an average of 0.49 mmHg ($p < 0.001$) and DBP decreased by 0.20 mmHg ($p < 0.001$).

When accounting for classification, the target group of stage 2 hypertension patients experienced a statistical significant decrease in blood pressure (Model 2). On average each month following their initial reading, their SBP decreased by 1.21 mmHg ($p < 0.001$), and diastolic blood pressure (DBP) decreased by 0.41 mmHg ($p < 0.001$). Stage 1 hypertension patients also saw statistically significant decreases, but the remaining groups of normal and elevated blood pressure did not experience a statistically significant decrease in blood pressure.

Figure 1 presents the marginal effects of the interaction term in Model 2 alongside the overall population change in blood pressure from Model 1. The graph demonstrates a significant monthly reduction in both SBP and DBP among the general patient population, with a steeper decline observed in stage 2 hypertension patients.

Results of sensitivity analysis

The sensitivity analysis revealed the unrestricted model, including additional predictors (i.e. race, age, hour of recording), did not significantly enhance the model's explanatory ability when compared to restricted model.

Discussion

In this study, we observed that the population, on average, experienced a decrease in SBP and DBP from their initial readings. Stratification by baseline blood pressure showed that patients in the stage 2 hypertension category achieved the most significant improvements. Collectively these findings suggest that the virtual companion was effective in reducing blood pressure across our study population, particularly among the target group of stage 2 Hypertensive patients, who saw the most notable decreases over the study's duration.

Our findings that the virtual companion reduced blood pressure is consistent with multiple randomized control trials. This aligns with findings reported in a similar study from Scotland, which documented an average systolic blood pressure reduction of 6.55 mmHg and a diastolic reduction of 4.23 mmHg over a span of 12 months.[7] In our study, conducted over a shorter duration, we observed an average systolic decrease of 5.21 mmHg and a diastolic decrease of 2.19 mmHg. While the study conducted in Scotland evaluated the broader impact of the intervention, it did not specify the reduction within each blood pressure classification. In contrast, our analysis revealed a more significant average reduction of 13.03 mmHg systolic and 4.39 mmHg diastolic blood pressure within the stage 2 Hypertesive class during our study period. Moreover, our study population was older on average, more closely reflecting the demographics typically affected by hypertension. When examining the influence of race, our analysis indicated that race did not have a statistically significant effect on the outcomes.

This study has several limitations. First, we relied on self-reported blood pressure measurements. This approach inherently carries the risk of recall bias, as patients are asked to input their usual blood pressure readings from the preceding week. Instead of capturing real-time data, the study had patients rely on memory, possibly leading to inaccuracies in the measurements recorded.

Nevertheless**,** this approach aligns with real-world hypertension management. While capturing multiple real-time readings weekly or using Bluetooth-connected devices would be ideal to reduce bias, it may not be practical for all patients.

The second limitation of this study was the unintended inclusion of many controlled normotensive patients in the program, due to broad enrollment criteria and lack of a baseline period. Although our hypertension registry was valuable to identify patients, the registry did not always reflect the patient's current condition, potentially including those with transient elevations like white-coat syndrome. The inclusion of normotensive patients inadvertently skewed the overall outcome, as these patients experienced minimal change values.

The enrollment challenges encountered in this study highlight the potential difficulties that can arise during the implementation of hypertensive interventions, suggesting issues that other groups may also face when launching similar programs.

In summary, we found that telemonitoring through a virtual companion is effective in reducing blood pressure for those with stage 2 hypertension. However, programmatic improvements could be made, particularly related to patient enrollment. Our findings suggest that implementing similar interventions could be a good option for managing hypertension, particularly in more severe cases, within broader healthcare settings.

Citations

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